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Mondrian Conformal Regressors with Overlapping Categories

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Abstract

Normalized conformal regressors have two potential limitations: i) they may generate prediction intervals that are several times larger than the largest observed absolute residual in the calibration set, and ii) the interval sizes may exhibit high dispersion even when using non-informative difficulty estimators. Mondrian conformal regressors provide a remedy by forming categories through equal-sized binning of the difficulty estimates and applying standard conformal regressors within each category. A drawback of this approach is that the mapping from difficulty estimates to prediction intervals may be very coarse, in particular for smaller calibration sets. Moreover, the calibration set size of each category cannot be fully controlled, potentially leading to overly conservative prediction intervals. We introduce a generalization of conformal predictors that allows overlapping Mondrian categories, and present an instantiation of this class for conformal regression. The proposed approach results in a smoother distribution of interval sizes while ensuring that each Mondrian category contains exactly the desired number of examples. A large-scale experimental evaluation on 33 datasets shows that the proposed approach consistently achieves superior average rankings with respect to prediction interval size compared with standard, normalized, and non-overlapping Mondrian conformal regressors over a range of difficulty estimators and significance levels.

Keywords: Conformal regressors, Mondrian conformal predictors, Mondrian conformal regressors.

1. Introduction

Conformal regression allows for turning point predictions into prediction intervals with a guaranteed coverage rate. The guarantee holds independently of the technique for generating the model that provides the point predictions, e.g., ridge regression (Papadopoulos et al., 2002), neural networks (Papadopoulos and Haralambous, 2010), k-nearest neighbors (Papadopoulos et al., 2011), and random forests (Johansson et al., 2014; Boström et al., 2017).

The size of a prediction interval may be seen as an indication of the uncertainty in a specific prediction. In the standard setting (resulting in so-called *standard conformal regressors*), the intervals are formed by subtracting and adding a quantile of the absolute residuals in the calibration set. Although valid, this results in intervals that are not adapted to the individual test objects. To get more informative, and also on average tighter, intervals, the absolute residuals may be normalized using an object-specific difficulty estimate, and where the lower and upper bound of the intervals are obtained by multiplying the difficulty of the test object to some quantile of the normalized calibration scores. Although more informative, intervals generated in this way may suffer from the following two problems, as pointed out in (Boström and Johansson, 2020); i) the intervals may be several times larger than the largest observed absolute residual (in contrast standard conformal regressors, for which the intervals may at most be two times the largest absolute residual in the calibration set), and ii) for a non-informative, e.g., completely random, difficulty estimator, there may still be a high dispersion of the sizes of the output intervals, giving a false impression of a high variance in uncertainty.

The solution proposed in (Boström and Johansson, 2020) was to form Mondrian categories by equal-sized binning of the difficulty estimates for the calibration set and generate a standard conformal regressor for each category. Since each conformal regressor is a standard conformal regressor formed from a subset of the calibration set, each output interval is bounded in size by twice the largest observed absolute residual. Moreover, if the difficulty estimator is completely random, the division into categories corresponds to a random partitioning, for which we should expect similar quantiles, and hence a limited dispersion of the interval sizes.

However, the formation of Mondrian categories by equal-sized binning has some potential drawbacks. The prediction intervals are generated using a step function with the undesired effect that instances with similar difficulty estimates that end up on different sides of a bin boundary may be assigned intervals of significantly different size; this thresholding effect is avoided when employing normalized conformal regressors as the interval sizes are directly proportional to the difficulty estimate. The problem is particularly noticeable for smaller calibration sets, which can be partitioned into only a few bins, resulting in a coarse mapping from difficulty estimates to prediction interval sizes. Another side-effect of forming Mondrian categories by binning is that it is often not possible to divide the calibration set in such a way that the number of instances in each category is equal to the number required for not producing overly conservative intervals, i.e., $n/\epsilon - 1$ for some $n = 1, 2, \ldots$ and significance level $\epsilon \in (0, 1)$. For example, if $\epsilon = 0.05$ then a category with 39 ($n = 2$) or 59 ($n = 3$) instances will result in intervals with exact coverage, while any calibration set size in between will result in intervals with a conservative coverage, as discussed in (Johansson et al., 2015).

In this work, we extend the Mondrian framework to allow for overlapping categories, i.e., each example may belong to more than one category. We show that if a test object is assigned to one of the categories randomly, the coverage guarantee still holds, under the usual assumption of exchangeability. Based on this we propose to apply Mondrian conformal regressors within this extended framework and form overlapping categories based on the ranks of the difficulty estimates. We show that the procedure indeed results in more smooth distributions of intervals, while still maintaining the bound of the interval sizes as provided by regular Mondrian conformal regressors, i.e., intervals with sizes that are more than twice the largest observed absolute residual will never be generated.

The main contributions of the paper are:

- Mondrian conformal predictors are extended to handle overlapping categories
- Mondrian conformal regressors that use multiple partitionings of the difficulty estimates are proposed

- A large-scale empirical investigation is presented comparing standard, normalized and Mondrian conformal regressors, with both non-overlapping and overlapping categories.

In the next section, we describe standard, normalized and Mondrian conformal regressors, together with the novel class of *Overlapping Conformal Predictors* and its adaptation to conformal regressors, i.e., *Overlapping Mondrian Conformal Regressors*. In Section 3, we describe experimental setup and results. Finally, we summarize the main findings and outline some directions for future work in Section 4.

2. Method

2.1. Conformal Predictors

Given a multiset of examples $\{z_1, \ldots, z_{q+1}\}$, where each $z_i \in \mathcal{Z}$, for some example space $\mathcal{Z}$, we may obtain a multiset of nonconformity scores $\{\alpha_1 = A(z_1), \ldots, \alpha_{q+1} = A(z_{q+1})\}$ using some nonconformity function $A : \mathcal{Z} \to \mathbb{R}$. The example space $\mathcal{Z}$ may consist of, e.g., objects and/or labels, and the employed nonconformity function A, which here is assumed to be fixed prior to observing the examples, is assumed to handle the examples accordingly.

From the multiset of nonconformity scores $\{\alpha_1, \ldots, \alpha_{q+1}\}$ and a value τ that is sampled uniformly from the unit interval, a (smoothed) p-value for the $q + 1$th nonconformity score is computed by:

$$p_{q+1} = \frac{|i : \alpha_i > \alpha_{q+1}| + \tau |i : \alpha_i = \alpha_{q+1}|}{q+1} \tag{1}$$

where $i \in \{1, \ldots, q+1\}$ (Vovk et al., 2022).

For prediction tasks, each example z_i consists of a pair $(\boldsymbol{x}_i, y_i)$, where $\boldsymbol{x}_i \in \mathcal{X}$, for some object space $\mathcal{X}$, and $y_i \in \mathcal{Y}$ for some label space $\mathcal{Y}$. For classification tasks, the label space is finite, i.e., $\mathcal{Y} = \{l_1, \ldots, l_k\}$, while for regression tasks, the label space typically consists of the real numbers, i.e., $\mathcal{Y} = \mathbb{R}$.

Given a calibration (multi-)set $Z_{cal} = \{z_1, \ldots, z_q\}$, a nonconformity function A, a significance level ϵ, and a test object $\boldsymbol{x}_{q+1}$, a *conformal predictor* outputs a prediction set:

$$C = \{l \in \mathcal{Y} : z_{q+1} = (\boldsymbol{x}_{q+1}, l) \ \& \ p_{q+1} \geq \epsilon\} \tag{2}$$

Let $z_{q+1} = (\boldsymbol{x}_{q+1}, y_{q+1})$ where y_{q+1} is the true target for the object $\boldsymbol{x}_{q+1}$. Under the assumption that the examples $\{z_1, \ldots, z_{q+1}\}$ are exchangeable, it follows that p_{q+1} is distributed uniformly on $[0, 1]$ (Vovk, 2021). This gives the following (exact) coverage guarantee:

$$\mathbb{P}(y_{q+1} \in C) = 1 - \epsilon \tag{3}$$

For classification, the prediction set can be obtained by straightforwardly enumerating the elements of the label space, while for regression, prediction sets (intervals) are typically obtained by relying on a specific format of the nonconformity function:

$$A((\boldsymbol{x}, y)) = \frac{|y - h(\boldsymbol{x})|}{\delta(\boldsymbol{x})} \tag{4}$$

where y is the correct label for the object $\boldsymbol{x}$, $h(\boldsymbol{x})$ is the predicted label using some model h and $\delta(\boldsymbol{x})$ is the estimated difficulty, using some *difficulty estimator* δ. A prediction interval can then be computed by:

$$P = [h(\boldsymbol{x}_{q+1}) - \delta(\boldsymbol{x}_{q+1})\, \alpha_{(j)}, h(\boldsymbol{x}_{q+1}) + \delta(\boldsymbol{x}_{q+1})\, \alpha_{(j)}] \tag{5}$$

where $\alpha_{(1)}, \ldots, \alpha_{(q)}$ denotes a permutation of $\alpha_1, \ldots, \alpha_q$ sorted in descending order and $j = \lfloor \epsilon\,(q{+}1) \rfloor$. It should be noted that this construction may result in conservative prediction intervals, and instead of a guarantee of exact coverage, we get a lower bound:

$$\mathbb{P}(y_{q+1} \in P) \geq 1 - \epsilon \tag{6}$$

2.2. Mondrian Conformal Predictors

Assume that we are given a function to map examples to a set of categories $\mathcal{M} = \{1, \ldots, c\}$ using a function (*Mondrian categorizer*) $M : \mathcal{Z} \to \mathcal{M}$. For prediction tasks, where each example $z_i = (\boldsymbol{x}_i, y_i)$, with $\boldsymbol{x}_i \in \mathcal{X}$ and $y_i \in \mathcal{Y}$, the categories may be formed using the labels, the objects or a combination of both. A Mondrian categorizer M can be used to partition the calibration set $Z_{cal} = \{z_1, \ldots, z_q\}$ into subsets $Z_1, \ldots, Z_c$, where each subset Z_k is defined by:

$$Z_k = \{z_i : z_i \in Z_{cal} \;\&\; M(z_i) = k\} \tag{7}$$

In this work, we will focus on a subclass of Mondrian categorizers, which we refer to as *object-based Mondrian categorizers*, that assigns categories based on the objects ($\boldsymbol{x}_i$) only.

An (object-based) *Mondrian conformal predictor* is a conformal predictor that forms a prediction set by using the subset of the calibration set that corresponds to the category that is assigned to the test object $\boldsymbol{x}_{q+1}$ instead of using the full calibration set, i.e., Z_k is used instead of Z_{cal}, where $k = M(\boldsymbol{x}_{q+1})$, when computing p_{q+1} (Eq. 1). The coverage guarantees (Eq. 3 and 6) still hold under the assumption of the exchangeability for $\{z_1, \ldots, z_{q+1}\}$, as exchangeability is implied for all subsets, including the subset that contains all examples that have been assigned to a specific category.

2.3. Overlapping Mondrian Conformal Predictors

In this work, we extend Mondrian conformal predictors to handle *Overlapping Mondrian categorizers* (denoted M^{++}), which map examples to any subset of the categories $\mathcal{M} = \{1, \ldots, c\}$, i.e., $M^{++} : \mathcal{Z} \to \mathcal{P}(M)$. Let $K \subseteq M$ denote the categories that have been assigned to a test example z_{q+1}. We refer to a conformal predictor that forms a prediction set by selecting uniformly at random one of the assigned categories, i.e., $k \sim \mathcal{U}(K)$, and using Z_k as the calibration set, as an *Overlapping Mondrian conformal predictor*.[1]

For a randomly selected category $k \in K$, exchangeability of $Z_k \cup \{z_{q+1}\}$ follows from the assumption of exchangeability of $Z_{cal} \cup \{z_{q+1}\}$, due to the former being a subset of the latter.

1. In case M^{++} outputs an empty set ($\emptyset$), this is handled as a separate category that contains all calibration examples.

2.4. Overlapping Mondrian Conformal Regressors

We introduce a novel approach to conformal regression, called Overlapping Mondrian Conformal Regressor (OMCR), which is a specific form of Overlapping Mondrian conformal predictor for regression that employs a difficulty estimator to form adaptive prediction intervals. The approach employs Alg. 1 to assign one of the overlapping categories to a test object. It returns the corresponding subset of the calibration set, which is given as input together with the test object, difficulty estimator and size of the category.

The algorithm computes the ranks of all objects based on their estimated difficulty, with a lower rank assigned to less difficult objects. If the test object is ranked among the bottom (or top) n objects, then a category corresponding to the n bottom-ranked (or top-ranked) objects are returned. If the test object is ranked in between of the n bottom- and top-ranked objects, a category is formed such that the rank of the test object within the category is uniformly distributed between 1 (top-ranked) and $n+1$ (bottom-ranked). This is equivalent to considering all categories of size $n+1$ formed from consecutive difficulty estimates (including the difficulty estimate for the test object) and randomly selecting one of these categories. The proposed approach is hence an instance of the class of Overlapping Mondrian conformal predictor and inherits its coverage guarantees.

In Fig. 1, the random selection of one of ten overlapping categories of size $n = 9$ is shown, where the rank of the difficulty estimate for the test object is 10 and the size of the calibration set is $q = 20$.

Algorithm 1: Difficulty-based Category Assignment

Input: $Z_{cal} = \{(\boldsymbol{x}_1, y_1), \ldots (\boldsymbol{x}_q, y_q)\}$ (calibration set),
$\boldsymbol{x}_{q+1}$ (test object),
δ (difficulty estimator),
n (category size)

$\Delta = \{\delta_1, \ldots, \delta_{q+1}\} \leftarrow \{\delta(\boldsymbol{x}_1), \ldots, \delta(\boldsymbol{x}_{q+1})\}$
$r_1, \ldots, r_{q+1} \leftarrow rank(\delta_1, \Delta), \ldots, rank(\delta_{q+1}, \Delta)$
if $r_{q+1} < n+1$ **then**
  $Z_k \leftarrow \{(\boldsymbol{x}_i, y_i) : i \in \{1, \ldots, q\} \; \& \; r_i \leq n+1\}$
else if $r_{q+1} > q-n$ **then**
  $Z_k \leftarrow \{(\boldsymbol{x}_i, y_i) : i \in \{1, \ldots, q\} \; \& \; r_i \geq q-n\}$
else
  $k \sim \mathcal{U}(1, n+1)$
  $Z_k \leftarrow \{(\boldsymbol{x}_i, y_i) : i \in \{1, \ldots, q\} \; \& \; r_{q+1-k} < r_i < r_{q+1}\} \cup$
    $\{(\boldsymbol{x}_i, y_i) : i \in \{1, \ldots, q\} \; \& \; r_{q+1} < r_i \leq r_{q+2+n-k}\}$
return Z_k

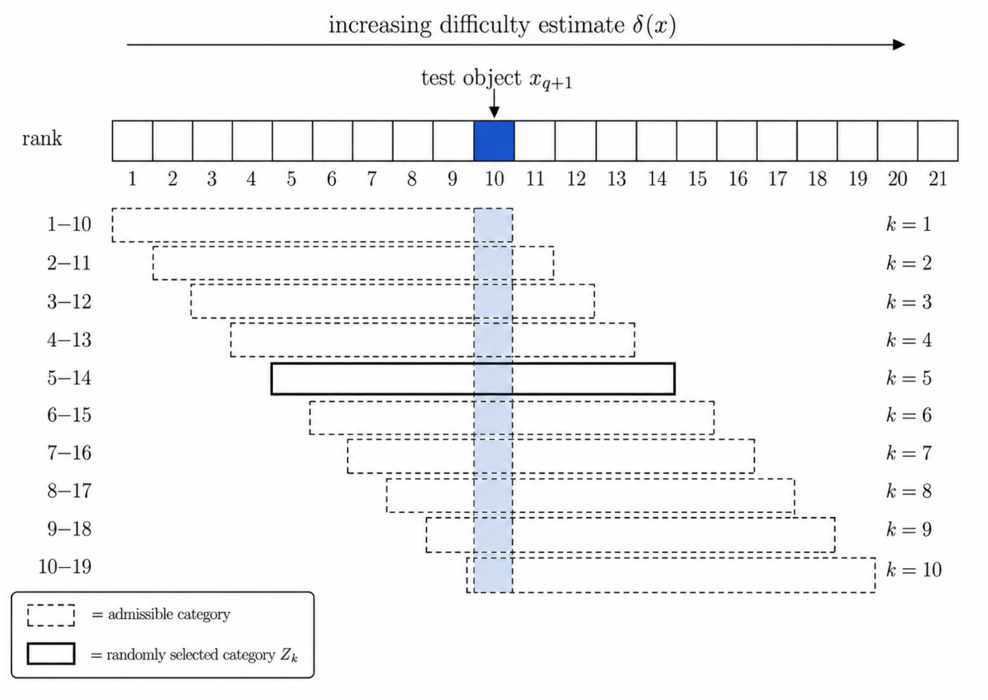


Figure 1: Random selection of one of 10 overlapping categories.

3. Experimental Evaluation

In this section, we will evaluate the efficiency (prediction interval size) of Overlapping Mondrian Conformal Regressors (OMCR) compared to standard, normalized and regular Mondrian conformal regressors. We will start with an illustrative example and then describe the setup and present results for the full-scale experiment.

3.1. Illustration

Let us first illustrate how the alternative approaches perform on the `house_sales` dataset from `openml`[2], containing house sale prices for King County (U.S) between May 2014 and May 2015, with 21 613 observations and 19 features. A random half of the data was used for testing, while the other half was randomly divided into a proper training set (80%) and a calibration set (20%). A `RandomForestRegressor` from `scikit-learn`,[3] with 500 trees was generated from the proper training set, which was applied to the calibration and test objects. A variance-based difficulty estimator, as implemented in `crepes`[4] was further employed, for the normalized and Mondrian conformal regressors. For the normalized conformal regressor, a positive term β has to be added to the to the difficulty estimates to avoid division by zero in Eq. 4, but also to control the size of the intervals. To narrow the range of feasible values for β, a min-max scaler was employed to the difficulty estimates, and results for three values

2. https://www.openml.org
3. https://scikit-learn.org
4. https://github.com/henrikbostrom/crepes/

of the parameter are explored here; $\beta \in \{0.1, 0.01, 0.0001\}$. It should be noted that β and min-max scaling has no effect on the standard and Mondrian-based approaches; the latter are only affected by transformations that change the ordering of the difficulty estimates. For the Overlapping Mondrian Conformal Regressor (OMCR), a category size of 199 instances was chosen, while the number of bins for the regular Mondrian conformal regressor was set to 10, giving an average category size of 216.2.

In Fig. 2, the empirical cumulative distributions of the interval sizes for the test set are shown for the four approaches, when $\epsilon = 0.05$, together with the three settings of β (affecting the normalized conformal regressor only). The standard approach, which does not employ a difficulty estimator, results in wider intervals for most predictions compared to any of the three informed approaches. It can be seen that the distribution of interval sizes for the regular Mondrian conformal regressor follows a step function, with one fixed interval size for each of the 10 categories. In contrast, despite forming categories of approximately the same size, the intervals output by the Overlapping Mondrian conformal regressor follow a clearly more smooth distribution. The distribution of interval sizes for the normalized conformal regressor is heavily dependent on β; a too high value (Fig. 2a) results in the smallest output intervals being much wider than the ones output by the Mondrian approaches, while a too low value (Fig. 2c) results in intervals that are more than two times wider than the largest observed absolute residual in the calibration set (1.3×10^6).

3.2. Experimental Setup

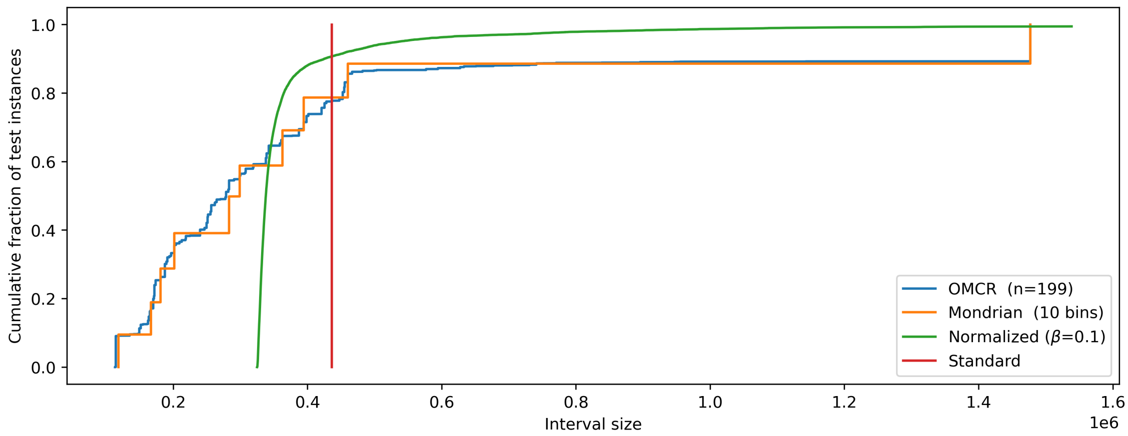
3.2.1. Datasets

For the full-scale experiment, we used 33 of the 35 datasets in the collection `OpenML-CTR23 - A curated tabular regression benchmarking suite` (`openml` collection id: 353), excluding two datasets with missing values. In Table 1, the names and `openml` ids of the datasets are shown, together with the number of instances, features and numerical features, before converting any categorical features to numerical by one-hot encoding.

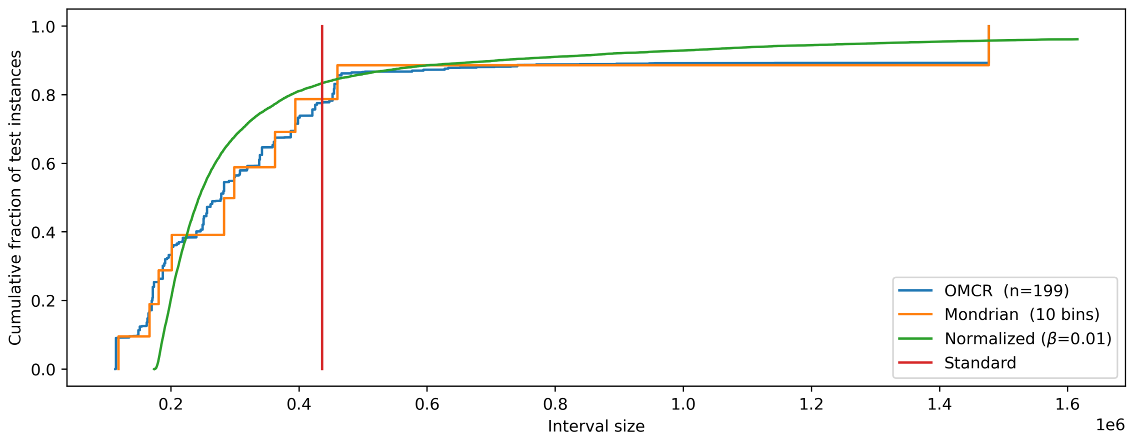
3.2.2. Test Protocol

Each dataset was used in conjunction with 10-fold cross-validation, where on each iteration, one fold was held out for testing and the remaining folds were concatenated and randomly split into a proper training set (70%) and a calibration set (30%). The median prediction interval size was recorded for each fold and averaged across the 10 folds for each conformal predictor, using three significance levels: $\epsilon \in \{0.1, 05, 0.01\}$.

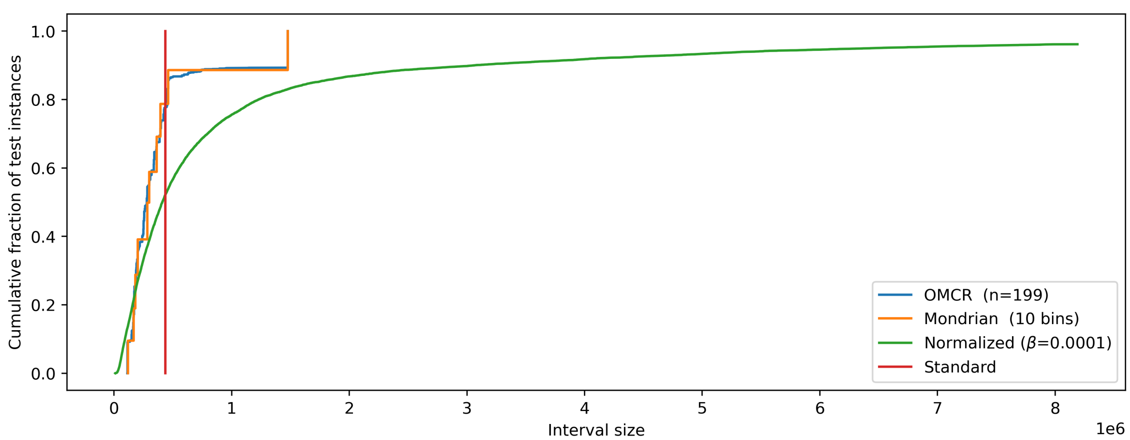
A `RandomForestRegressor` from `scikit-learn` with 500 trees was generated from each proper training set, and used to generate point predictions for the calibration and test objects. Three types of difficulty estimator were further employed; i) a variance-based difficulty estimator (`var`), which measures the variance of the predictions among the ensemble members (individual trees in the forest), ii) a k-nearest neighbor estimator (`knn`), which measures the variance among the labels of the $k = 25$ nearest neighbors in the proper training set, in min-max normalized Euclidean space, and iii) the point prediction itself (`pred`), i.e., assigning higher difficulty for higher predicted values. A min-max scaler was fitted to the difficulty estimates of the proper training set and used to scale the difficulty estimates of



(a) $\beta = 0.1$



(b) $\beta = 0.01$



(c) $\beta = 0.0001$

Figure 2: Empirical cumulative distributions of interval sizes on the `house_sales` dataset.

Table 1: Employed datasets from `OpenML-CTR23`.

Dataset	ID	#inst	#feat	#num. feat
abalone	44956	4177	8	7
airfoil_self_noise	44957	1 503	5	5
auction_verification	44958	2 043	7	5
concrete_compressive_strength	44959	1 030	8	8
physiochemical_protein	44963	45 730	9	9
superconductivity	44964	21 263	81	81
geographical_origin_of_music	44965	1 059	116	116
solar_flare	44966	1 066	10	2
naval_propulsion_plant	44969	11 934	14	14
white_wine	44971	4 898	11	11
red_wine	44972	1 599	11	11
grid_stability	44973	10 000	12	12
video_transcoding	44974	68 784	18	16
wave_energy	44975	72 000	48	48
sarcos	44976	48 933	21	21
california_housing	44977	20 640	8	8
cpu_activity	44978	8 192	21	21
diamonds	44979	53 940	9	6
kin8nm	44980	8 192	8	8
pumadyn32nh	44981	8 192	32	32
miami_housing	44983	13 932	15	15
cps88wages	44984	28 155	6	2
socmob	44987	1 156	5	1
kings_county	44989	21 613	21	17
brazilian_houses	44990	10 692	9	5
health_insurance	44993	22 272	11	4
fifa	45012	19 178	28	27
energy_efficiency	44960	768	8	8
forest_fires	44962	517	12	10
student_performance_por	44967	649	30	13
QSAR_fish_toxicity	44970	908	6	6
cars	44994	804	17	17
space_ga	45402	3 107	6	6

the calibration and test objects, clipping values below 0 and above 1. The value $\beta = 0.01$ was further added to the difficulty estimates.

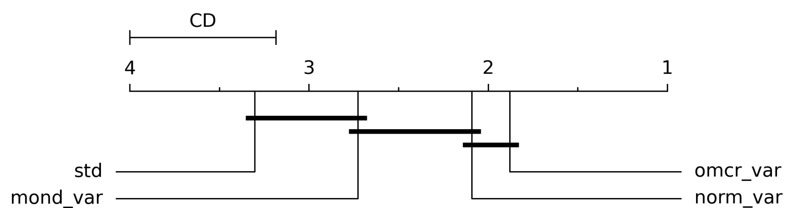
Four conformal regressors were formed and applied for each fold; i) a standard conformal regressor (ignoring the difficulty estimates), ii) a normalized conformal regressor (computing intervals as in Eq. 4), iii) a non-overlapping Mondrian conformal regressor, employing equal-sized binning of the difficulty estimates, and iv) an Overlapping Mondrian conformal regressor. The category sizes (n) for the latter were $n = 99$ for $\epsilon \in \{0.1, 0.05\}$ and $n = 199$ for $\epsilon = 0.01$. The number of bins for the (non-overlapping) Mondrian conformal regressor was set to match this, i.e., $\lfloor q/n \rfloor$, where q was the size of the calibration set. All algorithms and estimators, except for the Overlapping Mondrian conformal regressor, were obtained from `crepes`.

3.3. Experimental Results

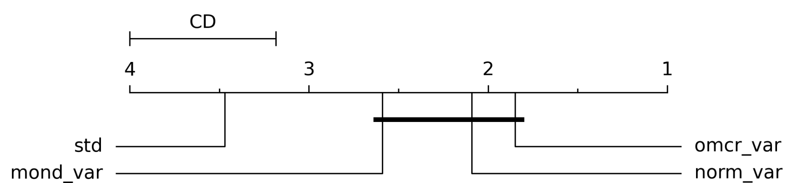
In Fig. 3, the average ranks according to prediction interval size (lower rank = smaller intervals) over the 33 datasets are shown for the four conformal regressors, when using the variance-based difficulty estimator. In Fig. 3a, the results for the significance level $\epsilon = 0.1$ shows that the proposed approach (OMCR) significantly (with 95% confidence) outperforms both the standard and non-overlapping Mondrian conformal regressors, according to a Friedman test, followed by a post-hoc Nemenyi test. For the lower levels of significance (Fig. 3b and Fig. 3c), the only remaining significant difference is that to the standard conformal regressor.

In Fig. 4, the average ranks in prediction interval size are shown for the knn-based difficulty estimator. It can be seen that the proposed approach (OMCR) significantly outperforms the normalized conformal regressor for all three levels of significance, and also the standard conformal regressor for the two smallest significance levels.

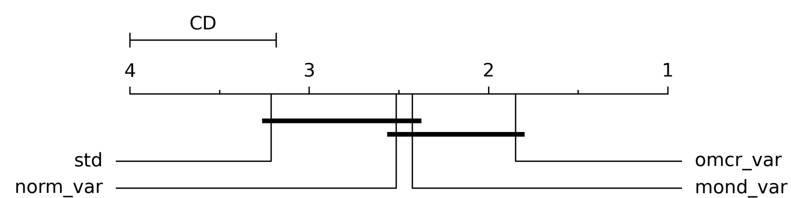
In Fig. 5, the average ranks are shown for when using the point predictor as a difficulty estimator, showing that the proposed approach (OMCR) significantly outperforms all competing approaches for $\epsilon = 0.1$ (Fig. 5a), the standard and normalized conformal regressors for $\epsilon = 0.05$ (Fig. 5b), and the normalized conformal regressor for $\epsilon = 0.01$ (Fig. 5c).



(a) $\epsilon = 0.1$

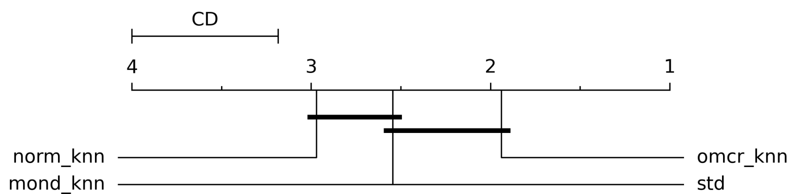


(b) $\epsilon = 0.05$

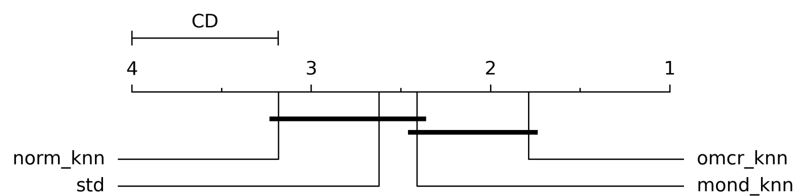


(c) $\epsilon = 0.01$

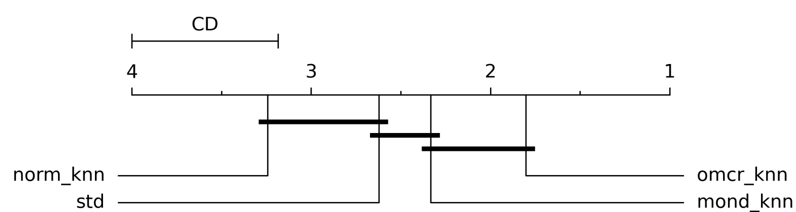
Figure 3: Average ranks when using the difficulty estimator `var`.



(a) $\epsilon = 0.1$

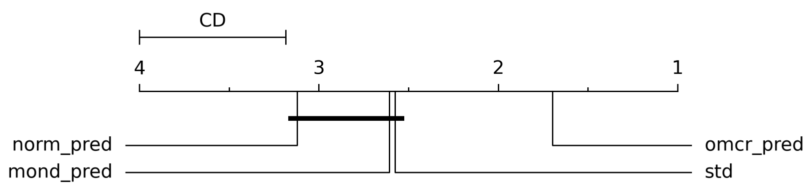


(b) $\epsilon = 0.05$

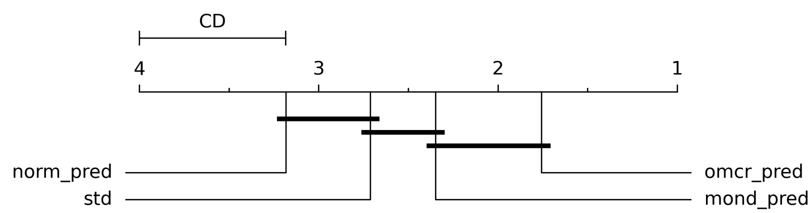


(c) $\epsilon = 0.01$

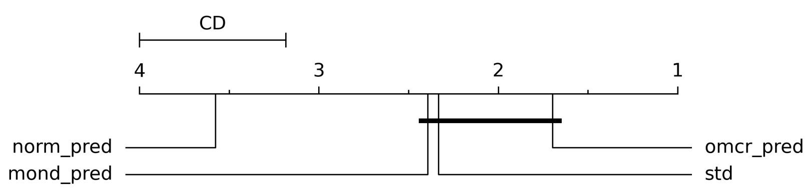
Figure 4: Average ranks when using the difficulty estimator `knn`.



(a) $\epsilon = 0.1$



(b) $\epsilon = 0.05$



(c) $\epsilon = 0.01$

Figure 5: Average ranks when using the difficulty estimator `pred`.

4. Concluding Remarks

We have introduced Overlapping Mondrian Conformal Predictors, which allows intersecting Mondrian categories. We proposed an instantiation of the framework to address two limitations of Mondrian conformal regressors, i.e., a potentially too coarse distribution of prediction interval sizes and conservative coverage due to the size of each category not being well-aligned with the chosen significance level. The novel approach, called Overlapping Mondrian Conformal Regressors, results in smoother distributions while maintaining the coverage guarantee and bound of the prediction interval size. By construction, the approach further guarantees that each category contains exactly the desired number of examples, thereby addressing the problem of overly conservative prediction intervals. Results from a large-scale experimental evaluation on 33 datasets show that the proposed approach compares favorably to standard, normalized and non-overlapping Mondrian conformal regressors in terms of efficiency (prediction interval size) across three different difficulty estimators and significance levels.

There are several directions for future research. One direction concerns exploring alternatives to selecting one of the overlapping categories uniformly at random, e.g., using some informed approach, while maintaining the guarantees. Another direction is to extend the experimental evaluation, by including other competing approaches, such as CQR (Romano et al., 2019) and deterministic versions of the proposed approach, alternative performance metrics, e.g., quantile (pinball) loss, and difficulty estimators. Finally, an interesting direction for future work is to apply the general framework of Overlapping Mondrian Conformal Predictors also to other type of conformal predictors, e.g., conformal predictive systems (Vovk et al., 2020; Boström et al., 2021).

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